

GPS Location Tracking Application Using MIT App Inventor

Abstract

The GPS Location Tracking Application is a simple and user-friendly Android application developed using MIT App Inventor. The main purpose of this application is to obtain the user's current geographical location using GPS and display it accurately on a map. The app uses built-in location services of the mobile device and integrates OpenStreetMap through the Map component. This application is useful for navigation, location identification, and learning GPS-based mobile application development.

Introduction

Global Positioning System (GPS) technology plays a vital role in modern mobile applications such as navigation, tracking, and location-based services. With the increasing use of smartphones, GPS applications have become essential in daily life.

This project demonstrates how a GPS-based application can be developed easily using **MIT App Inventor**, a visual programming platform. The application allows users to retrieve their current latitude and longitude and view their real-time location on a map with the click of a button.

Objectives of the Project

- To design a simple GPS-based Android application
- To retrieve real-time location using the device GPS
- To display the current location on a map
- To understand the use of LocationSensor and Map components
- To learn block-based programming using MIT App Inventor

Software Requirements

- MIT App Inventor (Web-based platform)
- Web Browser (Google Chrome recommended)
- Android Smartphone
- MIT AI2 Companion App

Hardware Requirements

- Android Mobile Phone with GPS support
- Internet Connection
- PC or Laptop

Tools & Technologies Used

- MIT App Inventor
- OpenStreetMap
- GPS (Location Sensor)
- Block-based Programming
- Android Platform

System Architecture

The application consists of three main components:

1. **User Interface** – Button and Map display
2. **Location Sensor** – Fetches latitude and longitude
3. **Map Component** – Displays the current location visually

When the user clicks the “Get Location” button, the app fetches GPS coordinates from the LocationSensor and moves the map view to the current position.

Application Components Description

1. Screen Component

- Acts as the main container for all UI elements
- Displays the map and button

2. Map Component

- Displays the geographical map using OpenStreetMap
- Shows the user’s current location
- Supports zooming and panning

3. Button Component

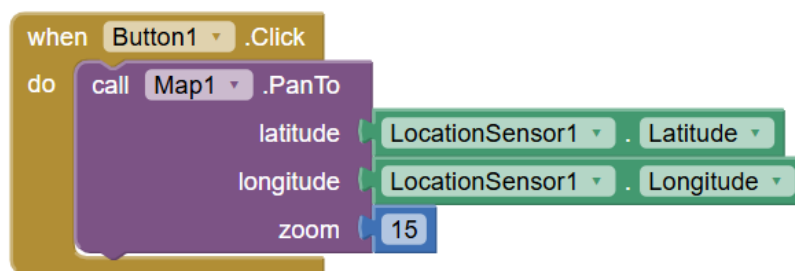
- Labeled as “**Get Location**”
- Used to trigger location fetching

4. LocationSensor Component

- Fetches real-time latitude and longitude
- Uses GPS and network services

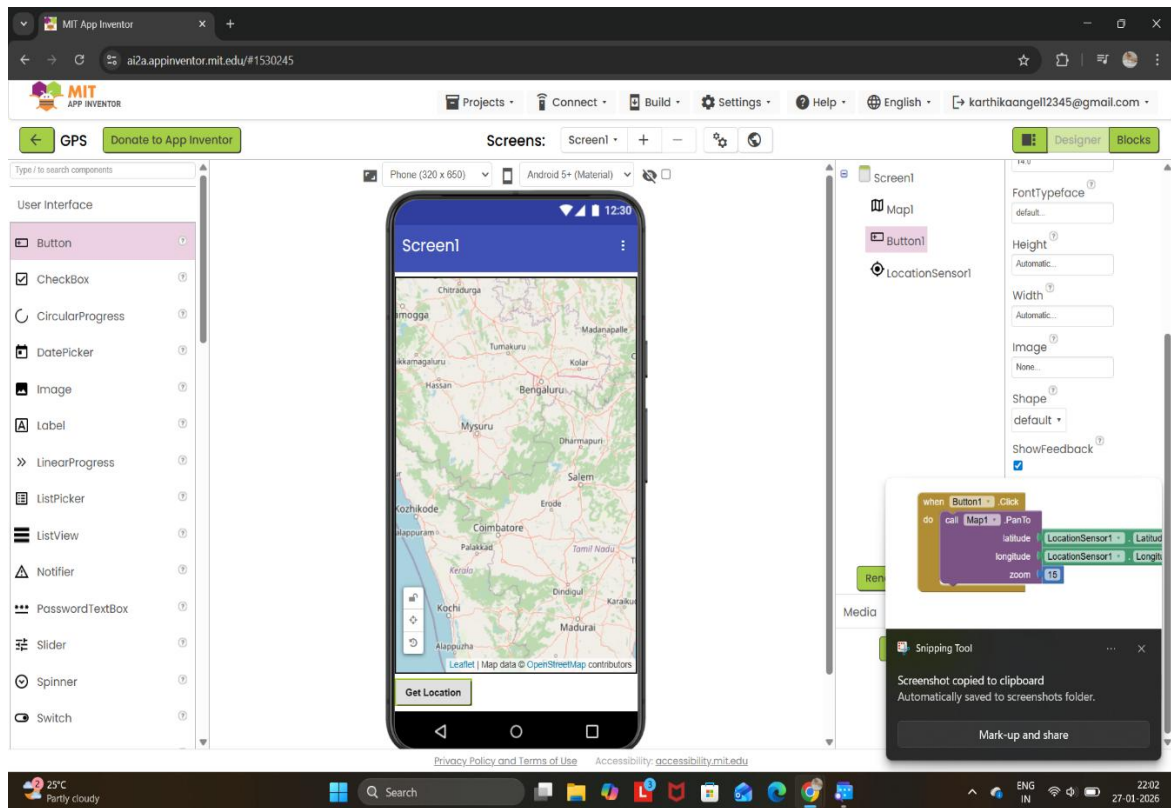
Block Diagram / Logic Description

- When the **Get Location** button is clicked
- The app reads **Latitude** and **Longitude** from LocationSensor
- The Map component pans to the received coordinates
- The zoom level is set for clear visibility



Working of the Application

1. The user opens the application
2. The map is displayed on the screen
3. The user clicks the **Get Location** button
4. GPS permission is requested (if not already granted)
5. The app retrieves current latitude and longitude
6. The map automatically moves to the user's location
7. The current position is clearly visible on the map



Advantages of the Application

- Easy to use
- Real-time location tracking
- No complex coding required
- Lightweight application
- Useful for beginners learning GPS apps

Limitations

- Requires internet connection for map loading
- GPS accuracy depends on device and signal strength
- Limited features compared to advanced navigation apps

Applications of the Project

- Location tracking
- Navigation assistance

- Educational learning tool
- Emergency location identification
- Field surveys

Future Enhancements

- Add marker to show exact position
- Display latitude and longitude values
- Add route navigation feature
- Store location history
- Integrate Google Maps

Conclusion

The GPS Location Tracking Application successfully demonstrates the implementation of location-based services using MIT App Inventor. The project helps in understanding GPS functionality, map integration, and block-based mobile application development. This application is efficient, simple, and suitable for beginners as well as educational purposes.

References

- MIT App Inventor Official Documentation
- OpenStreetMap Contributors
- Android GPS Documentation